The Lehmann Kallen form

The Analysis

In this section, we try to learn about the exact propagator $\Delta(x-y)$ from general principles.

$$\Delta(x-y) \equiv i\langle 0|T\varphi(x)\varphi(y)|0\rangle$$

We assume the field to be normalized such that,

$$\langle 0|\varphi(x)|0\rangle = 0 \quad ; \quad \langle k|\varphi(x)|0\rangle = e^{-ikx} \quad ; \quad \langle k|k'\rangle = (2\pi)^{d-1}\, 2\omega\, \delta^{d-1}(\vec{k}-\vec{k}')$$

The lorentz invariant phase-space differential is:

$$d\tilde{k} \equiv \frac{d^{d-1}k}{(2\pi)^{d-1}\,2\omega}$$

Now, using the completeness relation for all one particle eigenstates:

$$\Delta(x-y) = i\langle 0|T\varphi(x)\varphi(y)|0\rangle$$

$$= i\int d\tilde{k}\, \langle 0|\varphi(x)|k\rangle\langle k|\varphi(y)|0\rangle$$

$$= i\int \frac{d^{d-1}k}{(2\pi)^{d-1}\,2\omega}\, e^{ik(x-y)} \equiv \int \frac{d^dk}{(2\pi)^d}\, e^{ik(x-y)}\, \tilde{\Delta}(k^2) \longrightarrow \text{Exact momentum-space prop.}$$

In the free-field theory, the momentum space propagator is given by:

$$\tilde{\Delta}(k^2) = \frac{1}{k^2+m^2-i\epsilon}$$

Now that we have defined everything, lets start our analysis by inserting a complete set of energy eigenstates in the first equation.

Note that we have 3 types of energy eigenstates:

1. vacuum state $|0\rangle$
2. One particle states $|k\rangle \rightarrow \omega = (\vec{k}^2+m^2)^{\frac{1}{2}}$
3. many particle states, these are specified by the three mom^n $\vec{k}$ and a bunch of other parameters (termed n).

   $\omega = (\vec{k}^2+M^2)^{\frac{1}{2}}$, $M \geq 2m$, $M \in n$

Thus we have:

$$\langle 0|\varphi(x)\varphi(y)|0\rangle = \langle 0|\varphi(x)|0\rangle\langle 0|\varphi(y)|0\rangle + \int d\tilde{k}\, \langle 0|\varphi(x)|k\rangle\langle k|\varphi(y)|0\rangle$$

$$+ \sum_n \int d\tilde{k}\, \langle 0|\varphi(x)|k,n\rangle\langle k,n|\varphi(y)|0\rangle$$

We can handle the first two terms by the normalization of the field, for the second:

if we sub in $\varphi(x) = e^{-ip^\mu x_\mu} \varphi(0) e^{ip^\mu x_\mu}$ ↳ Energy-momentum operator.

We get:

$$\langle k,n|\varphi(x)|0\rangle = e^{-ikx}\langle k,n|\varphi(0)|0\rangle \qquad (k^0 = \omega)$$

This gives us:

$$\langle 0|\varphi(x)\varphi(y)|0\rangle = \int d\tilde{k}\, e^{ik(x-y)} + \sum_n \int d\tilde{k}\, e^{ik(x-y)} |\langle k,n|\varphi(0)|0\rangle|^2$$

We can take this further by defining a spectral density:

$$\rho(s) \equiv \sum_n |\langle k,n|\varphi(0)|0\rangle|^2\, \delta(s-m^2)$$

This gives us:

$$\langle 0|\varphi(x)\varphi(y)|0\rangle = \int d\tilde{k}\, e^{ik(x-y)} + \int_{4m^2}^{\infty} ds\, \rho(s) \int d\tilde{k}\, e^{ik(x-y)}$$

Swapping the x and y:

$$\langle 0|\varphi(y)\varphi(x)|0\rangle = \int d\tilde{k}\, e^{-ik(x-y)} + \int_{4m^2}^{\infty} ds\, \rho(s) \int d\tilde{k}\, e^{-ik(x-y)}$$

We can use both of these and:

$$\int \frac{d^dk}{(2\pi)^d} \frac{e^{ik(x-y)}}{k^2+m^2-i\epsilon} = i\theta(x^0-y^0)\int d\tilde{k}\, e^{ik(x-y)} + i\theta(y^0-x^0)\int d\tilde{k}\, e^{-ik(x-y)}$$

We can get:

$$i\langle 0|T\varphi(x)\varphi(y)|0\rangle = \int \frac{d^dk}{(2\pi)^d} e^{ik(x-y)} \left[\frac{1}{k^2+m^2-i\epsilon} + \int_{4m^2}^{\infty} \frac{1}{k^2+s-i\epsilon}\right]$$

With this we have the Lehmann-Kallen form of the exact momentum space propagator $\tilde{\Delta}(k^2)$:

$$\tilde{\Delta}(k^2) = \frac{1}{k^2+m^2-i\epsilon} + \int_{4m^2}^{\infty} ds\, \rho(s) \frac{1}{k^2+s-i\epsilon}$$